

Probability & Statistics Short Note

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1 Introduction to Statistics

- The science of collecting, organizing, analyzing, presenting data and finally drawing conclusions in the best possible way.

1.1 Basic Terms

- **Population** : The complete collection of individuals or objects that are of interest of study.
- **Sample** : A subset of population.
- **Observation** : A value that variable assumes for a single element.
- **Data** : The set of observations collected for the variable.
- **Parameter** : A numerical summary measure used to describe a characteristic of a population.
- **Statistic** : A numerical summary measure used to describe a characteristic of a sample.

1.2 Types of Variables

- An observable, measurable or recordable characteristic of interest to study. Basic unit of analysis.
- There are two type of variables.
 - **Categorical (Qualitative) variables**
 - * Variable having categories or classification that are not numerical in nature.
 - * Qualitative variables can be further divided into two as follows,
 - * **Ordinal variables**
 - * **Nominal variables**
 - **Numerical (Quantitative) variables**
 - * Variable that are numerical in nature.
 - * Quantitative variables can be further divided into two as follows,
 - * **Discrete variables** : There is a gap between two possible values.
 - * **Continuous variables** : There is not theoretical gap between two possible values.

1.3 Measure of Location

- **Minimum** : The first observation of the ordered data set.
- **Maximum** : The last observation of the ordered data set.
- **Measure of Center** :
 - A quality locates of the data set.
 - The most common measure of center are,
 - * **Mean**
 - The average value of the observation.
 - Mean is the **informal** midpoint of the data set. Data set may of may not be divided into two equal parts.
 - There can be different versions of means, Arithmetic Mean, Weighted Mean.
 - If each number is increased by a constant c, the mean is increased by c.
 - If each number is multiplied by a constant k, the mean is multiplied by k.

$$\text{Mean } (\bar{x}) = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n} = \frac{\sum_{i=1}^n x_i}{n}$$

* **Median**

- Median is the **formal** mid point of the data set. Data set is exactly divided into two equal parts.
- Half of the observation is smaller and other half is larger than median.
- Value will be equal to mean if and only if the distribution is symmetric.
- For odd number of items,

$$\text{Median (M)} = \frac{(n+1)^{\text{th}}}{2} \text{ value}$$

- For even number of items,

$$\text{Median (M)} = \frac{\left(\frac{n}{2}\right) + \left(\frac{n}{2} + 1\right)^{\text{th}}}{2} \text{ value}$$

* **Mode**

- Most frequent observation.
- There may be more than one mode.
- There may be a data set without a mode.
- Only measure that can be used with both categorical and numerical data.

* Mean vs. Median vs. Mode

- Mean and median will be equal if and only if the distribution is symmetric.
- In a left skewed graph mean is less than the median, while in a right skewed graph it is larger than median.
- Mean is very sensitive with **external values**. Therefore it is **not a resistant measure**.
- On the other hand median and mode are **resistant measures**, since they do not respond strongly to changes in a few observations, no matter how large these can be.

• **Quartiles :**

- A quantity that divides the data set into four equal parts.

* First Quartile (25%) : $Q_1 = \frac{1 * (n+1)^{\text{th}}}{4} \text{ value}$

* Second Quartile (50%) : $Q_2 = \frac{2 * (n+1)^{\text{th}}}{4} \text{ value}$

* Third Quartile (75%) : $Q_3 = \frac{3 * (n+1)^{\text{th}}}{4} \text{ value}$

• **Percentile :**

- A quantity that divides the data set into hundred equal parts.
- k% of the data is smaller and (100 - k)% of the data is larger than the kth percentile P_k .
- $Q_1 = P_{25}$, Median = $Q_2 = P_{50}$, and $Q_3 = P_{75}$.
- Calculating percentile,
 - * Rank n data values, from the smallest to largest.
 - * Calculate $\frac{nk}{100}$.
 - * If the result (a) is an integer then, P_k will be equal to the average of a^{th} value and $a+1^{\text{th}}$ value.
 - * If the result (a) is a fractional number then, P_k will be the $\lceil a \rceil^{\text{th}}$ value.

1.4 Measure of Spread (Dispersion / Variability)

- Measures how far each value is away from measures of center.
- Without measures of dispersed, measure of center may be misleading.
- Measures of dispersion should be,
 - a real number.
 - zero if all the data is identical.
 - increases as the data becomes more diverse.
 - cannot be less than zero.
- **Range (R)**
 - $R = \text{largest value} - \text{smallest value}$.
 - Simplest measure of dispersion.
 - Gives an indication of the absolute spread of distribution.
 - Depends only on two extreme values.
 - Not very informative and affected by outliers.
- **Inter Quartile Range (IQR)**
 - $IQR = Q_3 - Q_1$
 - The spread of 50% of the data.
 - Upper and lower 25% are discarded.
 - Not affected by outliers.
 - Not centered on the middle unless the distribution is symmetric.
 - Often used with median.
- **Variance (S^2 / σ^2 / $\text{Var}(X)$)**

$$S^2 = \frac{\sum (x_i - \bar{x})^2}{(n - 1)} = \frac{\sum x_i^2 - n\bar{x}^2}{(n - 1)}$$

- Numerically, the variance equals the average of several squared deviations from the mean.
- A data set which is highly variable will have a larger variance compared to a data set which is relatively homogenous.
- If each number is increased by a constant c , the variance remains unchanged.
- If each number is multiplied by a constant k , the variance is multiplied by k^2 .
- **Standard Deviation (S / σ)**
 - Positive value of the square root of the variance.
 - If each number is increased by a constant c , the standard deviation remains unchanged.
 - If each number is multiplied by a constant k , the standard deviation is multiplied by k .
- **Skewness**
 - One measure of skewness is the Pearson's coefficient of skewness.
 - Generally skewness can take any value between +3 and -3.

$$\text{Pearson's coefficient of skewness} = \frac{\text{Mean} - \text{Mode}}{\text{Standard Deviation}} = \frac{3 * (\text{Mean} - \text{Median})}{\text{Standard Deviation}}$$

2 Introduction to Probability

2.1 Basic Terms

- **Random Experiment** : An experiment in which there is some uncertainty about the outcome that will be observed.
- **Outcome** : The results of an observation of the experiment.
- **Outcome Space (Universal Set / Sample Space)** : Set of all possible outcomes of the experiment. (S)
- **Equally Likely Outcome** : If any outcome of a random experiment is as likely to occur as any other outcome then they are called equally likely outcomes.

2.2 Events

- Subset of the outcome space.
- **Equally Likely Events** : If any event is as likely to occur as any other event then they are called equally likely events.
- **Mutually Exclusive Events** : Two events A and B are said to be mutually exclusive (ir disjoint) if $A \cap B = \emptyset$
- **Exhaustive Events** : If two events A and B are such that $A \cup B = S$ and $P(A \cup B) = 1$.
- **Elementary Event** : Event with just one outcome.
- **Independent Events**
 - Events A and B are said to be independent if and only if one occur without being affected by the other.
 - $P(A \cap B) = P(A)P(B)$
 - Properties of independent events,
 - * If $P(A) > 0$ and $P(B) > 0$, then A and B are independent iff $P(A|B) = P(A)$ and $P(B|A) = P(B)$
 - * If A and B independent events then A and B' are also independent events.
 - Independence of more than two events,
 - * Three events, A, B, and C are mutually independent iff,
 - * $P(A \cap B) = P(A)P(B)$
 - * $P(A \cap C) = P(A)P(C)$
 - * $P(B \cap C) = P(B)P(C)$
 - * $P(A \cap B \cap C) = P(A)P(B)P(C)$

2.3 Probability Functions

- Probability is a function, which we denote by P. Like all functions, P has a domain which we denote by G. Since P assigns values to events, G is a set of events.
- A probability function P, defined on domain G, assigns to each event $A \in G$ a unique real number $P(A)$ in a way which satisfies the following axioms,
 - Axiom 1 : $P(A) \geq 0$ for any $A \in G$.
 - Axiom 2 : $P(\Omega) = 1$.
 - Axiom 3 : $P(A \cup B) = P(A) + P(B)$ whenever $A \cap B = \emptyset$, for any $A \in G$.
- Properties of P,
 - $P(A') = 1 - P(A)$
 - $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

2.4 Classical, Relative Frequency Definitions

- Symmetry Consideration
 - Physical symmetry sometimes enables us to easily assign probability values to certain events.
 - Such as rolling of a six sided die. Clearly each of the six elementary events are equally likely provided the die is properly constructed.
 - $P(A) = \frac{n(A)}{n(\Omega)}$
- Relative Frequency
 - Suppose a random experiment can be repeated and one event which can be observed is A. Then if we denote $f_r(A)$, the frequency of occurrence of A in r repetitions of the experiment we have the following.
 - $P(A) = \frac{f_r(A)}{r}$

2.5 Conditional Probability

- If A and B are two events, such that $P(A) \neq 0$ and $P(B) \neq 0$, then the probability of A given that B has already occurred is written as $P(A|B)$.

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \text{ for } P(B) > 0 \qquad P(B|A) = \frac{P(A \cap B)}{P(A)} \text{ for } P(A) > 0$$

$$P(A \cap B) = P(A|B)P(B) = P(A|B)P(A)$$

2.6 Law of Total Probability

- For two sets A and B,
$$P(A) = P(A \cap B) + P(A \cap B')$$
- By using the definition of conditional probability,
$$P(A) = P(A|B)P(B) + P(A|B')P(B')$$

$$\therefore P(A) = \sum_{i=1}^n P(A \cap B_i) = \sum_{i=1}^n P(A|B_i)P(B_i)$$

2.7 Bayes' Theorem

- Suppose that $A_1, A_2, A_3, \dots, A_n$ are mutually exclusive and exhaustive events and B is an arbitrary event. Then for $i=1,2,3,\dots,n$

$$\sum_{i=1}^n P(A_i|B) = \frac{P(B|A_i)P(A_i)}{P(B)} \text{ for } P(B) > 0$$

3 Random Variables and Expectation

3.1 Random Variables

- Many real-world experiments produce uncertain outcomes. To analyze them mathematically, we often convert outcomes into numbers.
- Let Ω be the outcome space corresponding to a random experiment. A function $X : \Omega \rightarrow \mathbb{R}$ is called a random variable.
- Functions of Random Variables,
 - It is sometimes useful to define random variable as functions of other random variables.
Ex: If X is a random variable, then so are $2X + 5$ and X^2 .

3.2 Discrete Random Variables

- The random variable X is a discrete random variable if the values taken by the random variable contains only a finite number of sample points.

Ex. A fair coin toss for three times. $\Omega = \{hhh, hht, hth, htt, thh, tht, tth, ttt\}$

Let X = no. of heads observed, Y = no. of tails observed, and $Z = XY$ and $P(H) = \frac{1}{3}$

ω	HHH	HHT	HTH	HTT	THH	THT	TTH	TTT
X	3	2	2	1	2	1	1	0
Y	0	1	1	2	1	2	2	3
Z	0	2	2	2	2	2	2	0
$P(\omega)$	$\frac{1}{27}$	$\frac{2}{27}$	$\frac{2}{27}$	$\frac{4}{27}$	$\frac{2}{27}$	$\frac{4}{27}$	$\frac{4}{27}$	$\frac{8}{27}$

$$\Omega_X = \Omega_Y = \{0, 1, 2, 3\}$$

$$\Omega_Z = \{0, 2\}$$

$$P_X(3) = P_Y(0) = P(HHH) = \left(\frac{1}{3}\right)^3 = \frac{1}{27}$$

$$P_X(2) = P_Y(1) = P(HHT) = P(HTH) = P(THH) = \left(\frac{1}{3}\right)^2 * \frac{2}{3} = \frac{2}{27}$$

$$P_X(1) = P_Y(2) = P(TTH) = P(THT) = P(HTT) = \left(\frac{2}{3}\right)^2 * \frac{1}{3} = \frac{4}{27}$$

$$P_X(0) = P_Y(3) = P(TTT) = \left(\frac{2}{3}\right)^3 = \frac{8}{27}$$

- Expectation = $E(X) = \sum_{\omega \in \Omega} (X(\omega) \cdot P(\omega))$
- Variance = $\sigma_X^2 = E[(X - \mu)^2] = \sum_{X \in \Omega_X} (X - \mu)^2 \cdot P(X)$

3.3 Continuous Random Variables

- Continuous Random Variables are the random variables which can take any value in a given interval.
- Those variables cannot be considered as discrete random variables since they cannot be placed one-to-one corresponding with counting numbers.

- **Probability Density Function (P.D.F.) :**

- The probability properties of a continuous random variable X are specified by its probability density function $f(X)$
- Properties of a Probability Density Function,
 - * $f(X) > 0$ for every $X \in \mathbb{R}$
 - * $\int_{-\infty}^{\infty} f(X)dX = 1$
- Probability of an event $(a < X \leq b)$ is found from the density function f as follows,

$$P(a < X \leq b) = \int_a^b f(X)dX$$

- Expectation = $E(X) = \int X.f(X)dx$
- Variance = $Var(X) = E(X^2) - E(X)^2 = \int X^2.f(X)dx - E(X)^2$

3.4 Expectation ($E(X)$)

- When the range Ω_X of a random variable $X : \Omega \rightarrow \mathbb{R}$ contains only a few points it is easy to absorb the entire probability distribution.
- However it is often the case that Ω_X is very large. In this case it is useful to use Expectation rather than mean.
- This quantity is also known as the mean of probability distribution of X or the expected value of X .
- The mean of the random variable X is the value of $E(X)$.
- Properties of $E(X)$,
 - $E(X) = \sum_{X \in \Omega_X} (X.P(X))$
 - $E(aX + b) = aE(X) + b$ where a and b are constants.
 - $E(X + Y) = E(X) + E(Y)$ when X and Y are random variables defined on the same outcome space.

3.5 Variance ($Var(X)$)

- The variance of the random variable X is the quantity,

$$\sigma_X^2 = E[(X - \mu)^2]$$

- Then σ_X is called the standard deviation of X . Both standard deviation and the variance can be used as measures of the spread of a probability distribution.
- Properties of $Var(X)$,
 - $Var(X) = E(X^2) - \mu^2 = E(X^2) - E(X)^2$
 - $Var(aX) = a^2 Var(X)$ where a is a constant.
 - $Var(X + Y) = Var(X - Y) = Var(X) + Var(Y)$ when X and Y are random variables defined on the same outcome space.

4 Discrete Probability Distributions

4.1 Binomial Distribution ($X \sim \text{Bin}(n, p)$)

- Consider an experiment with only two possible outcomes, one is considered as success and the other is considered as failure.
- A Binomial situation arises when n independent trials of the experiment are conducted.
Ex1: Toss a coin 6 times, considering obtaining a head on a single trial as the success.
Ex2: Roll a die 10 times, considering obtaining an even number as the success.
- The probability of an event will happen exactly k times in n independent trials is given by the following probability function. Here p is the probability of success in a single trial.

$$P(X = k) = {}^nC_k \cdot p^k \cdot (1 - p)^{n-k}$$

- Mean / Expectation ($E(X)$) = np
- Variance ($\text{Var}(X) / \sigma^2$) = npq where $q = (1 - p)$
- Standard Deviation (σ) = $\sqrt{npq}$

4.2 Poisson Distribution ($X \sim \text{Po}(\lambda)$)

- A discrete random variable X having P.D.F. of the form,

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

- where λ can take any positive value, is said to follow Poisson Distribution.
- If an event is randomly scattered in time/space and has mean number of occurrence λ in a given time/space and if X is the random variable the number of occurrences in the given interval, then $X \sim \text{Po}(\lambda)$.

Ex : Car accidents on a particular stretch of a road in one day.

- Mean / Expectation ($E(X)$) = λ
- Variance ($\text{Var}(X) / \sigma^2$) = λ
- Standard Deviation (σ) = $\sqrt{\lambda}$

4.3 Discrete Uniform Distribution ($X \sim \text{U}(k)$)

- Equal assignment to all outcomes. i.e. all outcomes are equally likely.
- Discrete uniform distribution deals with isolated outcomes (independent) like rolling a fair die.
- Random variable is defined in a sample space $S = \{1, 2, 3, 4, \dots, k\}$

$$P(X = x) = \frac{1}{k}$$

- Mean / Expectation ($E(X)$) = $\frac{k+1}{2}$
- Variance ($\text{Var}(X) / \sigma^2$) = $\frac{k^2 - 1}{12}$
- Standard Deviation (σ) = $\sqrt{\frac{k^2 - 1}{12}}$

4.4 Geometric Distribution ($X \sim \text{Geo}(p)$)

- The Geometric Distribution models the number of independent Bernoulli trials needed to get the very first success.
- Unlike in the Binomial distribution where number of trials is fixed, in geometric distribution, the number of trials X is random. (Keep going until the first success)
- Let p be the probability of success.

$$P(X = x) = p(1 - p)^{x-1}$$

- Mean / Expectation ($E(X)$) = $\frac{1}{p}$
- Variance ($\text{Var}(X) / \sigma^2$) = $\frac{(1 - p)}{p^2}$
- Standard Deviation (σ) = $\sqrt{\frac{(1 - p)}{p^2}}$

4.5 Negative Binomial Distribution ($X \sim \text{NB}(k, p)$)

- The Negative Binomial Distribution is the generalization of the Geometric Distribution.
- While geometric distribution only counts the trials until the first success, negative binomial distribution counts upto the k^{th} success.
- Let p be the probability of success.

$$P(X = n) = {}^{(n-1)}C_{(k-1)} p^k (1 - p)^{n-k}$$

- Geometric distribution is a special case of negative binomial distribution when $k = 1$.
- Mean / Expectation ($E(X)$) = $\frac{k}{p}$
- Variance ($\text{Var}(X) / \sigma^2$) = $\frac{k(1 - p)}{p^2}$
- Standard Deviation (σ) = $\sqrt{\frac{k(1 - p)}{p^2}}$

4.6 Hypergeometric Distribution ($X \sim \text{Hypergeometric}(N, k, n)$)

- The Hypergeometric distribution models sampling without replacement from finite population.
- Unlike the binomial distribution, where trials are independent and probability of success stays constant, hypergeometric sampling changes the probability of success with every turn since trials are dependent.
- Let the sample space size be n in a population of size N with k successes and $N - k$ failures.

$$P(X = x) = \frac{{}^k C_x {}^{(N-k)} C_{(n-x)}}{{}^N C_n}$$

- Mean / Expectation ($E(X)$) = $n \frac{k}{N}$
- Variance ($\text{Var}(X) / \sigma^2$) = $n \left(\frac{k}{N} \right) \left(1 - \frac{k}{N} \right) \left(\frac{N - n}{N - 1} \right)$
- Standard Deviation (σ) = $\sqrt{n \left(\frac{k}{N} \right) \left(1 - \frac{k}{N} \right) \left(\frac{N - n}{N - 1} \right)}$

4.7 Summary of Discrete Distributions

	P.D.F.	Mean (μ)	Variance (σ^2)
Binomial Distribution	${}^nC_k p^k (1-p)^{n-k}$	np	npq
Poisson Distribution	$\frac{e^{-\lambda} \lambda^x}{x!}$	λ	λ
Uniform Distribution	$\frac{1}{k}$	$\frac{k+1}{2}$	$\frac{k^2-1}{12}$
Geometric Distribution	$p(1-p)^{x-1}$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
Negative Binomial Distribution	${}^{(n-1)}C_{(k-1)} p^k (1-p)^{n-k}$	$\frac{k}{p}$	$\frac{k(1-p)}{p^2}$
Hypergeometric Distribution	$\frac{{}^kC_x {}^{(N-K)}C_{(n-x)}}{{}^NC_n}$	$n \frac{k}{N}$	$n \left(\frac{k}{N} \right) \left(1 - \frac{k}{N} \right) \left(\frac{N-n}{N-1} \right)$

5 Continuous Probability Distributions

5.1 Normal Distribution ($X \sim N(\mu, \sigma^2)$)

- Normal Distribution is the most important continuous distribution in statistics. Many measured quantities in the natural science follow a normal distribution.

Ex: Heights, weights, and blood pressure readings across large human populations.

- A continuous random variable with mean μ and variance σ^2 which follows normal distribution has a P.D.F. as follows,

$$f(X) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(X-\mu)^2}{2\sigma^2}}$$

- The distribution is bell shaped and symmetrical about $X = \mu$.
- Mean / Expectation ($E(X)$) = μ
- Variance ($\text{Var}(X)$ / σ^2) = σ^2
- Standard Deviation (σ) = σ
- Properties of Normal Distribution,
 - If there are two independent random variables that follow normal distribution such that, $X \sim N(\mu_1, \sigma_1^2)$ and $Y \sim N(\mu_2, \sigma_2^2)$, then

$$X + Y \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$$

$$X - Y \sim N(\mu_1 - \mu_2, \sigma_1^2 + \sigma_2^2)$$
 - If a random variable that follows normal distribution such that, $X \sim N(\mu, \sigma^2)$, is multiplied by a constant,

$$aX \sim N(a\mu, a^2\sigma^2)$$

5.2 Standard Normal Distribution ($Z \sim N(0, 1)$)

- Consider the random variable Z where $Z = \frac{X - \mu}{\sigma}$
- Now if the random variable X has a normal distribution with mean μ and variance σ^2 , then the random variable Z has a standard normal distribution with mean 0 and variance 1.

$$\phi(X) = \frac{1}{\sqrt{2\pi}} e^{-\frac{X^2}{2}}$$

- Mean / Expectation ($E(X)$) = 0
- Variance ($\text{Var}(X) / \sigma^2$) = 1
- Standard Deviation (σ) = 1

5.3 Continuous Uniform Distribution ($X \sim U(a, b)$)

- Equal assignment to all outcomes. i.e. all outcomes are equally likely.
- Continuous uniform distribution deals with continuous measurements like time, height, etc.

$$f(x) = \frac{1}{b - a}$$

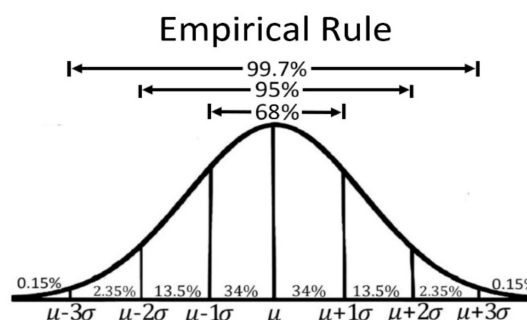
for $a \leq x \leq b$.

- Mean / Expectation ($E(X)$) = $\frac{a + b}{2}$
- Variance ($\text{Var}(X) / \sigma^2$) = $\frac{(b - a)^2}{12}$
- Standard Deviation (σ) = $\sqrt{\frac{(b - a)^2}{12}}$

5.4 Empirical Rule

- The Empirical Rule gives us how much of data lies within one, two or three standard deviations of the mean.

- Area under the total curve is 1.
- Approximately 68% of the distribution lies within $\pm 1\sigma$ of the mean.
- Approximately 95% of the distribution lies within $\pm 2\sigma$ of the mean.
- Approximately 99.7% of the distribution lies within $\pm 3\sigma$ of the mean.



- Since the normal distribution is perfectly symmetric, we can separate one, two and three standard deviation areas into buckets.
- From one standard deviation,

$$\frac{68\%}{2} = 34\%$$

$\therefore$ 34% of data lies between $(\mu, \mu \pm 1\sigma)$

- From two standard deviation,

$$\frac{95\% - 68\%}{2} = \frac{27\%}{2} = 13.5\%$$

$\therefore$ 13.5% of data lies between $(\mu \pm 1\sigma, \mu \pm 2\sigma)$

- From three standard deviation,

$$\frac{99.7\% - 95\%}{2} = \frac{4.7\%}{2} = 2.35\%$$

$\therefore$ 2.35% of data lies between $(\mu \pm 2\sigma, \mu \pm 3\sigma)$

5.5 Summary of Continuous Distributions

	P.D.F.	Mean (μ)	Variance (σ^2)
Normal Distribution	$\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{(X-\mu)^2}{2\sigma^2}}$	μ	σ^2
Standard Normal Distribution	$\frac{1}{\sqrt{2\pi}}e^{-\frac{X^2}{2}}$	0	1
Uniform Distribution	$\frac{1}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$

6 Distribution Approximation

6.1 Continuity Correction

- Continuity correction is used when approximation happens from a discrete random variable distribution to continuous random variable distribution.
- Continuity Correction Rules,
 - $P(X = k) \rightarrow P(k - 0.5 < Y < k + 0.5)$
 - $P(X \leq k) \rightarrow P(Y \leq k + 0.5)$
 - $P(X < k) \rightarrow P(Y \leq k - 0.5)$
 - $P(X \geq k) \rightarrow P(Y \geq k - 0.5)$
 - $P(X > k) \rightarrow P(Y \geq k + 0.5)$

6.2 Normal Approximation to Binomial Distribution

- If $X \sim \text{Bin}(n, p)$, then $\mu = np$ and $\sigma^2 = npq$.
- Now for larger distributions, where $n \geq 10$ and p is close to 0.5 or $n \geq 30$ and p is far from 0.5, we can use the normal approximation.
- $X \sim N(np, npq)$

6.3 Normal Approximation to Poisson Distribution

- If $X \sim \text{Po}(\lambda)$, then $\mu = \lambda$ and $\sigma^2 = \lambda$.
- Now for larger λ we can use the normal approximation.
- $X \sim N(\lambda, \lambda)$

7 Sampling Distribution and Central Limit Theorem

7.1 Population Distribution

- Population distribution is the probability distribution of random variable X in a population.
- It describes how the values of X are spread among individuals in the population

Ex: If X = height of all UCSC 1st year students, plotting all heights gives the population distribution. It may or may not be Normal.

- For a population with Mean = μ and Variance = σ^2 , $E(X) = \mu$ and $\text{Var}(X) = \sigma^2$. If it is Normal then $X \sim N(\mu, \sigma^2)$.

7.2 Sample Distribution

- A sample is n observation obtained from a population.
- Sample distribution is the distribution of n values in the sample.

Ex: Take $n = 50$ students, plot their 50 heights. That is the sample distribution. For large n , it resembles population distribution.

7.3 Sampling Distribution of Sample Mean $\bar{X}$

- Sampling distribution is the distribution of statistics.
- For sample mean $\bar{X}$, it is the distribution of $\bar{X}$ over all the possible samples of size n .
- This is done by repeatedly taking samples of size n , calculating their mean and plotting that $\bar{X}$.

Ex: Take 100 groups of 50 students, calculate mean height of each group. Plotting those 100 means gives the sampling distribution of $\bar{X}$.

- It is NOT the same as population distribution. It is much narrower.
- Let population mean = μ , variance = σ^2 and sample size = n . Then the sample mean will be $\bar{X} = \frac{\sum X_i}{n}$.
- Mean of sampling distribution : $\mu_{\bar{X}} = E(\bar{X}) = \mu$.
- Variance of sampling distribution : $\sigma_{\bar{X}}^2 = \text{Var}(\bar{X}) = \frac{\sigma^2}{n}$.
- Standard Error :

– Standard error of a sampling distribution is the standard deviation of the sampling distribution.

$$- \therefore \text{SE} = \sqrt{\text{Var}(\bar{X})} = \sqrt{\frac{\sigma^2}{n}} = \frac{\sigma}{\sqrt{n}}$$

7.4 Central Limit Theorem

- Central Limit Theorem used to determine the shape of a sampling distribution based on its sample size.
- The theorem suggests that if the size of the sample n is large enough, then the sampling distribution of the sample mean will be approximately normal no matter what the population distribution looks like.
- The sampling distribution of the sample mean will be approximately normal when the sample size is greater than or equal 30. (Rule of Thumb)
- Then by CLT, $X \sim N(\mu, \frac{\sigma^2}{n})$ or $Z = \frac{X - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$